AI Interview Technical Assessment System

Project Proposal Document

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1. Executive Summary

The **AI Interview Technical Assessment System** is a comprehensive, web-based, multi-tenant SaaS platform designed to streamline and automate the technical hiring process.

The system enables organizations to efficiently create, manage, and evaluate candidate assessments using AI-driven evaluation, real-time monitoring, and secure proctoring.

This solution is built to:

- Reduce manual screening effort
- Improve evaluation accuracy
- Enable scalable hiring across multiple organizations
- Deliver data-driven insights for faster decision-making

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2. Objectives

- Automate end-to-end candidate assessment workflows
- Standardize evaluation across roles and experience levels
- Enable AI-assisted scoring for subjective and coding responses
- Ensure test integrity through proctoring

- Provide actionable analytics and reporting for HR teams

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3. Platform Overview

The system is designed as a **multi-tenant SaaS platform**, allowing multiple organizations to operate independently within a shared infrastructure.

Each organization will have:

- Its own users (HR/Admin)
- Independent assessments
- Configurable policies
- Secure and isolated data

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4. Scope of Work

4.1 Candidate Management

- Automated email invitations with secure assessment links
- OTP-based authentication
- Attempt limits (max 2 attempts per candidate)
- Session management with reconnection support
- Real-time status tracking

4.2 Assessment Management

- Create assessments based on job roles
- Reusable assessment templates
- Metadata configuration:
    - Role / Title
    - Experience level
    - Skills

    ◦ Time limits
- Unique assessment link generation per candidate

4.3 Question Engine

Support for 11 question types:

1. MCQ
2. Multi-select MCQ
3. Short Answer
4. Code Completion
5. Coding Challenge
6. Debug & Fix
7. Code Review
8. SQL Query
9. System Design
10. Scenario / Case Study
11. Output Prediction

4.4 Coding Console

- In-browser code editor (IDE-like experience)
- Multi-language support (UI level)
- Syntax highlighting
- Auto-save functionality
- Code submission for evaluation

⚠️ **Note:** Real-time compilation/execution is not included in this phase.

4.5 AI-Powered Evaluation

Descriptive Evaluation

- Semantic answer analysis

- Contextual scoring
- Partial marking
- Confidence scoring

Coding Evaluation

- Logic and approach analysis
- Code quality feedback
- Pattern-based correctness detection

SQL Evaluation

- Query validation
- Output comparison
- Partial scoring

Smart Scoring Engine

- Unified scoring system
- Weighted evaluation
- Normalized ranking

AI Feedback System

- Performance reports
- Skill-wise insights
- Strengths and improvement areas

4.6 Proctoring & Monitoring

- Screen recording
- Camera and microphone capture
- Tab-switch detection
- External device detection (best effort)
- Violation logging

4.7 Analytics & Tracking

- Real-time event tracking:
    - Link opened
    - Assessment started
    - Submission completed
- HR dashboards
- Performance reports
- Audit trail

4.8 Role-Based Access Control

- Admin / HR roles
- Role-based visibility of results
- Access control per organization

4.9 Link & Session Management

- Configurable link expiry (default: 72 hours)
- Revoke and extend functionality
- Follow-up reminder system
- Single active session enforcement

4.10 Security

- Data encryption
- API validation
- Audit logs
- Tenant-level data isolation

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5. Multi-Tenant SaaS Architecture

The system will be implemented as a multi-tenant architecture with:

- Tenant-level data isolation using tenant_id
- Organization-specific configurations
- Secure access control per tenant
- Scalable infrastructure to support multiple organizations

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6. Technology Stack

Frontend

- React.js (TypeScript)
- Tailwind CSS / Material UI

Backend

- Node.js with NestJS

Database

- PostgreSQL

Real-Time

- WebSockets (Socket.io / NestJS Gateway)

AI Integration

- LLM APIs (e.g., OpenAI)

Proctoring

- WebRTC
- MediaRecorder API
- Screen Capture API

Code Editor

- Monaco Editor

Infrastructure

- AWS / Cloud hosting
- S3 (media storage)
- Redis (queue & caching)

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7. System Architecture (High-Level)

- Frontend (Candidate & HR UI)
- Backend API Layer
- Assessment Engine
- AI Evaluation Engine
- Proctoring Module
- Analytics Engine
- Cloud Storage

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8. Workflow Summary

1. HR creates/selects assessment
2. Candidate receives invitation link
3. Candidate logs in via OTP
4. Assessment starts with proctoring
5. System tracks activity in real time
6. Candidate submits responses
7. AI evaluates answers
8. HR reviews analytics and reports

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9. Development Effort Estimation

Total Development Hours

Scope	Hours

Optimized Full System	**940 – 1,160 hours**

Estimated Timeline

- Development Duration: **10 – 14 weeks**
- Milestone-based delivery

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10. Cost Estimation

At $6/hour

Scope	Cost
Minimum	**$5,640**
Maximum	**$6,960**

In INR (₹558/hour)

Scope	Cost
Minimum	**₹5,24,000 (approx)**
Maximum	**₹6,47,000 (approx)**

Note

Costs exclude:

- AI API usage
- Email services
- Cloud hosting and infrastructure

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11. Milestone-Based Delivery Plan

Milestone 1 – 10%

- Project setup
- Architecture design
- Documentation

Milestone 2 – 25%

- Candidate management
- Assessment setup
- Authentication

Milestone 3 – 25%

- Question engine
- Test platform
- Session handling

Milestone 4 – 30%

- AI evaluation
- Proctoring system
- Analytics dashboard

Milestone 5 – 10%

- Testing and optimization
- Deployment and handover

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12. Assumptions & Constraints

- Email-based candidate identification
- Asynchronous assessments
- English-only interface
- AI is assistive, not final decision-maker

- Recording retention: 90 days
- Uptime target: 99.9%

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13. Benefits

- Faster hiring process
- Reduced manual screening
- Improved evaluation accuracy
- Scalable hiring platform
- Better candidate experience

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14. Future Enhancements

- Live coding compiler integration
- Mobile application
- Advanced AI cheating detection
- Multi-language support
- ATS/HRMS integration

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15. Conclusion

The proposed **AI Interview Technical Assessment System** is a scalable and intelligent solution tailored for modern hiring needs.

By combining automation, AI capabilities, and secure proctoring, the platform enables organizations to conduct efficient, fair, and reliable technical assessments at scale.
